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Case Study

Rare case of mucopolysaccharidosis type II - HUNTER SYNDROME presenting with cardiac manifestation

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Abstract

The mucopolysaccharidoses (MPSs) are inherited lysosomal storage disorders caused by the absence of functional enzymes that contribute to the degradation of glycosaminoglycans (GAGs). The progressive systemic deposition of GAGs results in multi-organ system dysfunction. Cardiac involvement has been reported in all MPS syndromes and is a common and early feature, particularly for those with MPS I, II, and VI. Cardiac valve thickening with associated dysfunction has been reported 57% of patients with MPS II. Here we report a case of mucopolysaccharidosis type II who presented with cardiac mitralvalvularinvolvement.

Keywords: MPS: mucopolysaccharidosis, glycosaminoglycans (GAGs)

Introduction

The mucopolysaccharidoses (MPSs) are inherited lysosomal storage disorders caused by the absence of functional enzymes that contribute to the degradation of glycosaminoglycans (GAGs). The progressive systemic deposition of GAGs results in multi-organ system dysfunction

The mucopolysaccharide storage diseases express themselves clinically with a wide variety of abnormalities, including growth and mental retardation, skeletal abnormalities, clouded corneas, nerve compression syndromes, upper airway obstruction and cardiovascular involvement, to name the most common.

Severe cardiac involvement is a common feature of mucopolysaccharidoses (MPS), but occurs only rarely in MPS III (Sanfilippo syndrome). Cardiac involvement has

been reported in all MPS syndromes and is a common and early feature, particularly for those with MPS I, II, and VI.

Progressive cardiac valve pathology is the most prominent and uniform cardiac manifestation (60-90%) of patients with MPS1.

Other cardiac abnormalities include coronary artery disease, increased wall thickness and diffuse narrowing of the thoracic and abdominal aorta.

Cardiac valve thickening with associated dysfunction has been reported 57% of patients with MPS II2.

Here we report a case of mucopolysaccharidosis type II who presented with cardiac mitralvalvularinvolvement.

Case report

A 4 years old boy presented with. He was the first child born of a non consanguineous and healthy parents. Gestation and delivery were normal. The weight and length at birth were 2700 g and 46 cm respectively. Presenting complaints was recurrent respiratory tract infection, breathlessness NYHA class II and grunting sound. On physical examination, he had coarse facial features, with thick hair, a low hairline, thick eyebrows ,a wide and flat nasal bridge, thick lips and full cheeks, protruding tongue ,abdominal distension ,generalized hirsutism, short neck, andhepatosplenomegaly. No limitation of joint movement was present. Neurological examination showed clumsiness of fine movements. Mental retardation was also present.

An echocardiogram showed dilated cardiomyopathy, thickened anterior and posterior mitral leaflet with severe mitral insufficiency, and mild tricuspid insufficiency. Aortic valve was tricuspid and thickened. Ophthalmic examination was normal. Abdominal CT scan confirmed hepatosplenomegaly. The diagnosis of MPS(Hunter syndrome) was confirmed by decreased levels of idouronate 2 sulfatase enzyme on dry blood sample. The patient is planned for mitral valve repair /replacement.

Discussion:

Most studies have reported that valvular regurgitation, rather than stenosis, is more common and that the mitral valve, rather than the aortic, is most commonly affected. In general, left-sided valves (mitral & aortic) are more severely affected than those on the right side of the heart (tricuspid & pulmonary)3.

Conduction abnormalities and sinus tachycardia have been reported in 7% of a subset of the Hunter Outcome Survey of MPS II patients, andcomplete atrioventricular block (MPS II, III and VI) with subsequent need for cardiac pacing and sudden death have been reported4.

It is very important to recognize and treat this group of disorders as they carry a rapid downhill course .It is also essential to screen the family members for the

genetic mutation. Bone marrow transplantation is promising in certain types of mucopolysaccharidoses if diagnosed earlier.

References

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Figures



Photos of the boy to show physical features consistent with hunter syndrome (pic 1)

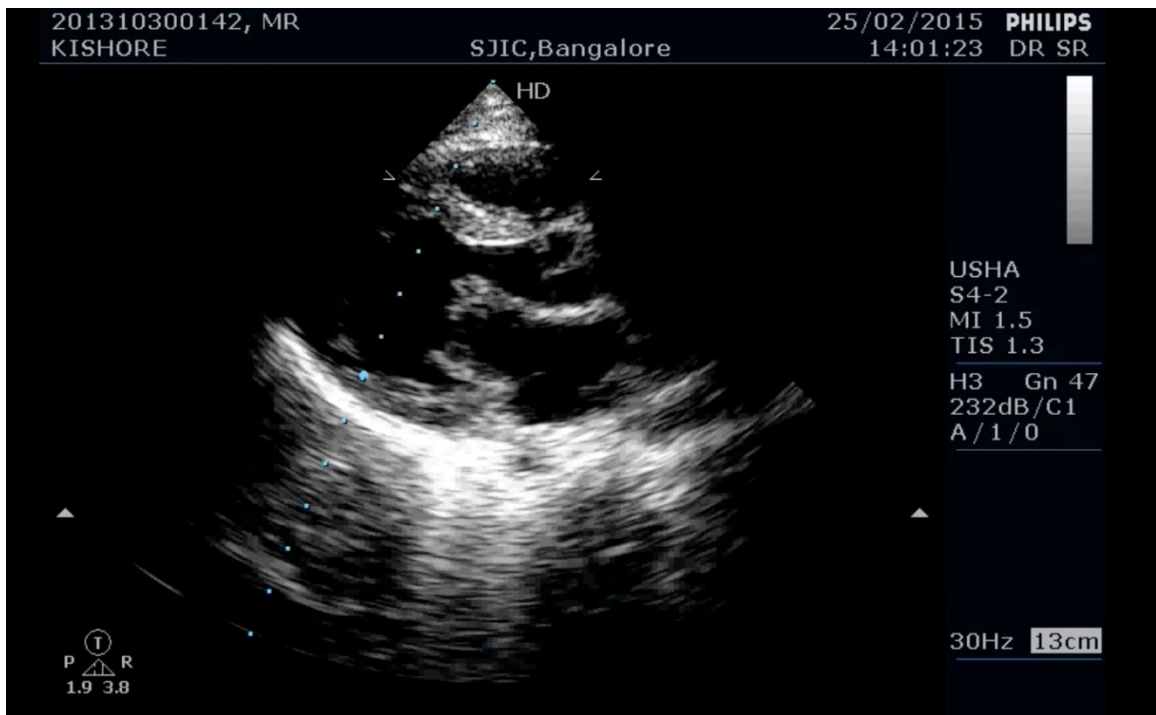


Fig 1 plax view showing grossly thickened mitral valve

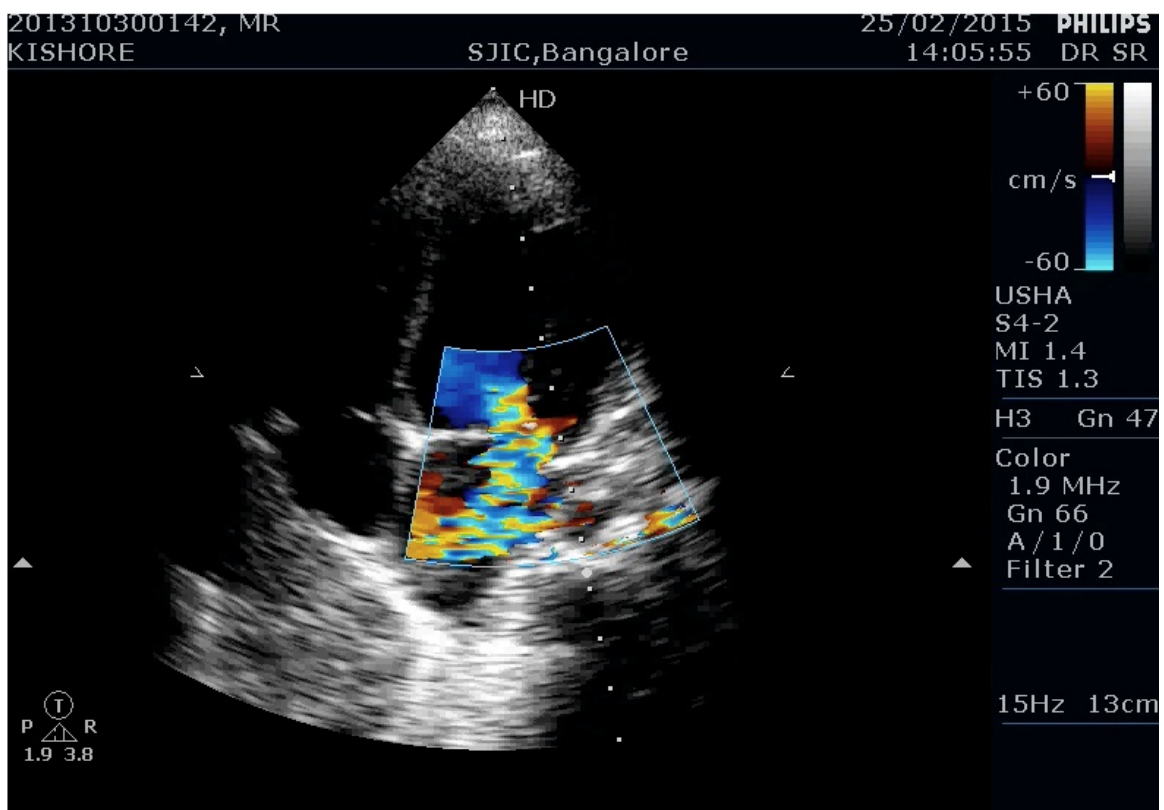


Fig 2 apical 4 chamber view showing thickened mitral valve with aml prolapse and severe mitral regurgitation.